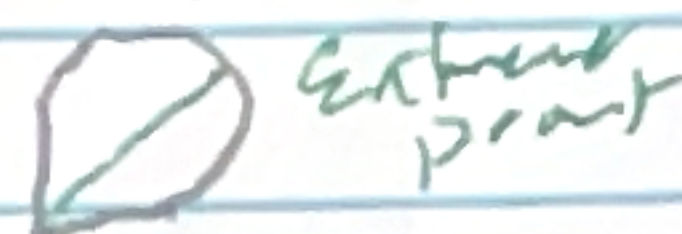


1) a. Region not convex

b. Region is convex. Can draw line @ any 2 points & stay in square.

Each of the 4 corners of square are extreme points

c. Region is convex. Line between any 2 points won't exit shape. Extreme point is between bottom left corner and top right on circle.



2) Basic s. 1 2: Let $x_3, x_4 = 0 \dots$

$$\begin{cases} x_1 + 3x_2 = 30 \\ 2x_1 + x_2 = 15 \end{cases}$$

$$\begin{aligned} \rightarrow x_1 + 3x_2 &= 30 \\ 6x_1 + 3x_2 &= 45 \end{aligned}$$

$$5x_1 = 15 \quad x_1 = 3$$

$$3 + 3x_2 = 30$$

$$3x_2 = 27 \quad x_2 = 9$$

$$(x_1, x_2, x_3, x_4) = (3, 9, 0, 0)$$

$$\text{Basic Matrix} \rightarrow \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$$

2. (cont.) Basic Sol 2: Let $x_1, x_3 = 0$

$$\begin{aligned}x_1 - x_3 &= 30 \\ 2x_1 + 2x_3 &= 15\end{aligned}$$

$$\rightarrow x_1 = 30 + x_3$$

$$\rightarrow 2(30 + x_3) + 2x_3 = 15$$

$$60 + 2x_3 + 2x_3 = 15$$

$$4x_3 = -45$$

$$x_3 = -\frac{45}{4}$$

$$\rightarrow x_1 = 30 + \left(-\frac{45}{4}\right)$$

$$x_1 = \frac{75}{4}$$

$$(x_1, x_2, x_3, x_4) = \left(\frac{75}{4}, 0, -\frac{45}{4}, 0\right)$$

$$\text{Basic Matrix} \rightarrow \begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix}$$

$$3.) \text{ a. } \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \\ 3 \end{bmatrix}$$

$$\rightarrow x_1 = 3 \quad (\text{Row 3})$$

$$\text{Sub } x_1 \text{ in R2: } 2(3) + 0(x_2) + 1(x_3) = 2$$

$$3 + x_3 = 2$$

$$\rightarrow x_3 = -1$$

3a (cont.) Sub x_1, x_5 in R1: $2(3) + 1(x_2) + 0(-1) = 7$

$$6 + x_2 = 7$$

$$\rightarrow x_2 = 1$$

$$(x_1, x_2, x_3, x_4, x_5) = (3, 0, 1, 0, -1)$$

b. Sol is not feasible, $x_5 < 0$ (where $x_5 = -1$)

4.) a. Given $(x_1, x_2) = (4, 10)$

$\rightarrow$ we can observe from graph that at point $S(4, 10)$

$$x_4 = 0$$

$$x_5 = 0$$

$\rightarrow$ Using first equation:

$$x_3 = 4 - (4 - 20)$$

$$\Rightarrow x_3 = 20$$

$$(x_1, x_2, x_3, x_4, x_5) = (4, 10, 20, 0, 0)$$

$$\text{Basic Matrix: } \begin{bmatrix} 1 & -2 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

b. Using basic vars x_1, x_2, x_4 . Solving $Bx = b$,

$$\text{Picking } B = \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \text{ to use } (-2)$$

$$b = \begin{bmatrix} 4 \\ 10 \\ 10 \end{bmatrix}$$

$$Bx_B = b : \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 18 \\ 10 \end{bmatrix}$$

From R3: $x_2 = 10$

Sub into R1: $x_1 - 2(10) = 4$
 $x_1 - 20 = 4$
 $\Rightarrow x_1 = 24$

Sub to R2:

$$2(24) + 10 + x_4 = 18$$

$$58 + x_4 = 18$$

$$\Rightarrow x_4 = -40$$

So,

$$x_B = \begin{bmatrix} x_1 \\ x_2 \\ x_4 \end{bmatrix} = \begin{bmatrix} 24 \\ 10 \\ -40 \end{bmatrix}$$

Since $x_4 = -40 \not\geq 0$, sol. is not feasible.